Individual assignment · Verification across optimization models

100 raw points · Individual submission

Deadline and course-grade weighting will be announced by the instructor. This assignment is a stand-alone problem set, not a replacement for the computational project.

Submit one PDF report with numbered answers and one runnable notebook or Python script. Include mathematical work, software versions, solver status, objective values and independent feasibility checks. A screenshot of a solver answer alone is insufficient.

You may consult course materials and documentation. Explain your own reasoning, cite external assistance and disclose any AI or code assistance. Do not submit another student's derivations or results. Aim for 5–8 report pages plus code.

1. LP certificate and sensitivity · 25 points

Maximize $4x+3y$ subject to $x+y\le5$, $2x+y\le8$ and $x,y\ge0$.

(5 points) Form the dual using nonnegative resource prices u,v.

(8 points) Find primal and dual optima. Verify feasibility, equality of objective values and every complementary-slackness product.

(6 points) Change only the first RHS to b_1. Derive the full interval where both resource constraints remain active and the same dual prices certify optimality.

(6 points) Predict and verify the optimal profit at $b_1=5.5$. Explain why extrapolating the same price to $b_1=9$ fails.

2. Integer production and start-up cost · 20 points

Maximize $7a+10b-2z$ subject to $2a+3b\le10$, $a\le4z$, a,b nonnegative integers, and z binary. Quantities a,b are batches and 2 is a fixed start-up cost.

(5 points) Explain the implication $a\le4z$ and whether $z=1$ forces $a>0$.

(5 points) Find the integer optimum using complete bounded enumeration. Include the best result with $z=0$ and $z=1$.

(5 points) Solve the LP relaxation, state its bound direction and compare it with the integer optimum.

(5 points) Give a feasible integer incumbent and the relaxation bound. State the interval containing the integer optimum before using enumeration to close the certificate.

3. Unconstrained iterations and Armijo · 20 points

Minimize $f(u,v)=(u-1)^2+2(v+1)^2$ from $r_0=(3,1)$.

(5 points) Compute gradient and Hessian. Classify the global minimizer.

(5 points) Take one negative-gradient step with fixed $\alpha=0.1$. Report the new point and objective.

(5 points) Restart from r_0. Apply Armijo with $c_1=0.1$, $\rho=0.5$ and initial $\alpha=1$. Show every trial until the first acceptance.

(5 points) Continue backtracking to a gradient norm below 10^{-8}. Submit a trace with current point, gradient, accepted α, objective and trial count. Explain why this trace differs from keeping $\alpha=0.1$ fixed.

4. Constrained NLP and a capacity decision · 35 points

Minimize $F(x,y)=(x-4)^2+2(y-3)^2$ subject to $x+y\le b$ and $x,y\ge 0$. Initially $b=4$. Use $g_1=x+y-b$, $g_2=-x$, $g_3=-y$ and $L=F+\Sigma\lambda_i g_i$.

(8 points) Derive the candidate with $x,y>0$ and active capacity. Compute all multipliers and F at $b=4$.

(8 points) Verify every KKT condition and explain whether it certifies a global optimum.

(7 points) Derive the b interval where this positive-coordinate active set applies. Treat endpoints separately.

(7 points) Compare $b=4$ with $b=4.5$. Calculate the exact objective improvement, its first-order prediction and the difference. If extra capacity costs 3.8 per capacity unit, should the plant buy the proposed 0.5-unit increment? Treat the quoted objective and capacity cost as comparable monetary amounts.

(5 points) Use an NLP solver from at least three distinct starting points. Report success/message, objective and feasibility/stationarity/complementarity residuals. Compare with your analytic solution.

Submission checklist

All four parts are answered. Figures label axes and units. Code runs from top to bottom. Results are compared with analytic calculations. Sensitivity conclusions state the parameter change and valid interval. Attach source code and any data you created.